

Songwen Hu

Ph.D. Student

College of Computing, School of Interactive Computing
Georgia Institute of Technology

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Area of Expertise

Data Visualization, Human Perception, Vision Science, Human-Centered AI, Recommendation System, Multimodal Interaction

Education

Georgia Institute of Technology

Ph.D. in Computer Science

Atlanta, GA, USA

Aug 2023 – Jun 2028 (Expected)

Advisor: **Dr. Cindy Xiong Bearfield**

Georgia Institute of Technology

M.S. in Computer Science, Visual Analytics

Atlanta, GA, USA

Aug 2023 – May 2026

Shanghai Jiao Tong University

B.Eng. in Electrical and Computer Engineering

Shanghai, China

Sep 2019 – Aug 2023

Advisor: **Dr. Shuai Li**

Teaching Experience

Course Instructor

Introduction to Information Visualization (CS 4460)

Georgia Tech

Summer 2026

Teaching Assistant

Introduction to Information Visualization (CS 4460)

Georgia Tech

Fall 2025

Industry Experience

Bosch China

Embedded Software Engineer Intern

Shanghai, China

Jan 2022 – Jun 2022

- Developed gesture recognition algorithms for human-vehicle interaction using DNNs.
- Applied attention mechanisms to the neural network for dynamic gesture classification.
- Achieved 90% accuracy for 16 static gestures and 9 dynamic gestures on webcam with 720p@30fps.

Grants and Gift Funds

Coco Grant (\$1,000)

Predict Reader Insights in Visualization by Modeling Cognitive Affordances

PI: Songwen Hu Faculty Advisors: Dr. Cindy Xiong Bearfield, Dr. Apurva Ratan Murty

MathWorks Gift Fund (\$2,000)

Gramm Extension for Exporting Function Compatibility

Working on MATLAB's open-source visualization library Gramm to support web interactivity.

Publications

2026

Hu, S., Hearst, M., Stokes, C., Xiong Bearfield, C. (2026). Visual Embellishments are Potential Distractions in Double-Column Reading. *To be presented at IEEE VIS 2026*.

Hu, S., Endert, A., Xiong Bearfield, C. (2026). Interaction Modulates Visual Interpretation and Trust in Uncertainty Visualizations. *Vision Science Society (VSS) 2026*.

2025

Hu, S., Yu, T., Rossi, R. A., Kim, S., Xiong Bearfield, C. (2025). VisChatter: Enhance Synchronous Collaboration on Visualization Dashboard through Visual Annotations. *Annual Meeting of the Cognitive Science Society (CogSci 2025)*.

Hu, S., Rossi, R. A., Yu, T., Wu, J., Zhao, H., Kim, S., Li, S. (2025). Interactive Visualization Recommendation with Hier-SUCB. *Proceedings of the International World Wide Web Conference (WWW 2025)*. [\[arXiv\]](#)

2024

Hu, S., Jiang, O., Riedmiller, J., Xiong Bearfield, C. (2024). Perceptual Benefits of Animation are Task-Dependent: Effects of Staging and Tracing in Dynamic Displays. *IEEE Transactions on Visualization and Computer Graphics, IEEE VIS 2024*. (Also presented as a poster at *Vision Science Society (VSS) 2024*.) [\[arXiv\]](#)

2022

Zuo, J., **Hu, S.**, Li, S., Yu, T., Zhao, H., Joe-Wong, C. (2022). Hierarchical Conversational Preference Elicitation with Bandit Feedback. *Proceedings of the Conference on Information and Knowledge Management (CIKM 2022)*. [\[arXiv\]](#)

Research Projects

Reverse-Engineering Insight Formation from Visualizations using Brain and Eye Signals

Advisors: Cindy Xiong Bearfield (Georgia Tech), Ratan Murty (Georgia Tech)

- Investigating how low-level perceptual features interact with high-level reasoning processes in visualization comprehension.
- Combining fMRI and eye-tracking to quantify perceptual and cognitive load across visual cortex and frontal cortex.
- Developing a computational model to predict and interpret perceptual–cognitive demands from visualizations.

VisChatter – Visual Annotations for Collaborative Analytics with Multimodality

Advisors: Cindy Xiong Bearfield (Georgia Tech), Tong Yu (Adobe)

- Designed and implemented an interactive visualization dashboard with real-time annotation for synchronous collaborative analytics.
- Integrated speech recognition, LLM-based keyword extraction, and custom annotation APIs for seamless multimodal input.
- Conducted controlled A/B testing, measuring user engagement and task efficiency improvements.

Interactive Web Integration for Gramm Visualizations in MATLAB

Advisors: Cindy Xiong Bearfield (Georgia Tech), Vijay Iyer (MathWorks), Pierre Morel (ULCO)

- Extended Gramm, an open-source MATLAB visualization library, with a feature to export plots as interactive web visualizations.
- Integrated Vega and SVG-based interactivity to enable zooming, panning, and tooltips directly in the browser.
- Collaborated with the open-source community to design, implement, and document the new functionality.

Assessing Immersiveness of Visual Storytelling in VR with Bio-signals

Advisors: Jeffery Riedmiller (Dolby Research), Cindy Xiong Bearfield (Georgia Tech)

- Developed a pipeline to collect and analyze physiological signals during VR storytelling.
- Piloting VR scenes varying in narrative structure and visual intensity.
- Conducting data-driven evaluation of perceived immersion and emotional response.

Vision Rehabilitation through Multisensory Interaction

Advisors: Tina Liu (Georgetown University), Cindy Xiong Bearfield (Georgia Tech)

- Built interactive prototypes for training spatial attention and visual reconstruction.
- Integrated visual and auditory cues to enhance sensory compensation and engagement.
- Piloting test sessions with clinical collaborators to assess user feedback and accessibility.

Assessing Cognitive Load of Word-scale Visualizations

Advisors: Marti Hearst (UC Berkley), Cindy Xiong Bearfield (Georgia Tech)

- Designed and executed a controlled eye-tracking study to assess how embedded visual embellishments affect reading comprehension across varied data densities.
- Quantified cognitive disruption by analyzing psychophysiological metrics (eye saccades).
- Explored word-scale visualization design guidelines for fluent reading flow.

Technical Skills

Programming & ML Frameworks: C++, MATLAB, JavaScript, Python, PyTorch, TensorFlow, R

Human-Computer Interaction: Interactive visualization (D3.js, Vega-Lite, Tableau), user study design, A/B testing

Data Science & Front-End Development: SQL (PostgreSQL, MySQL), Apache Spark, AWS, feature engineering, React

Natural Language Processing: Prompt engineering, LLM fine-tuning, Ollama, vLLM, LLM agents (LangGraph)